



Designing Your Capstone Infra Blueprint

Your first draft of an end-to-end architecture capturing the full lifecycle:
data → training → deployment → monitoring → governance

We'll refine this iteratively over Weeks 49–52 to create a solid foundation for your Capstone project.

Why a Blueprint Matters

A well-designed infrastructure blueprint:

Clear Roadmap

Provides a structured path for your Capstone build, reducing uncertainty and wasted effort

Early Gap Detection

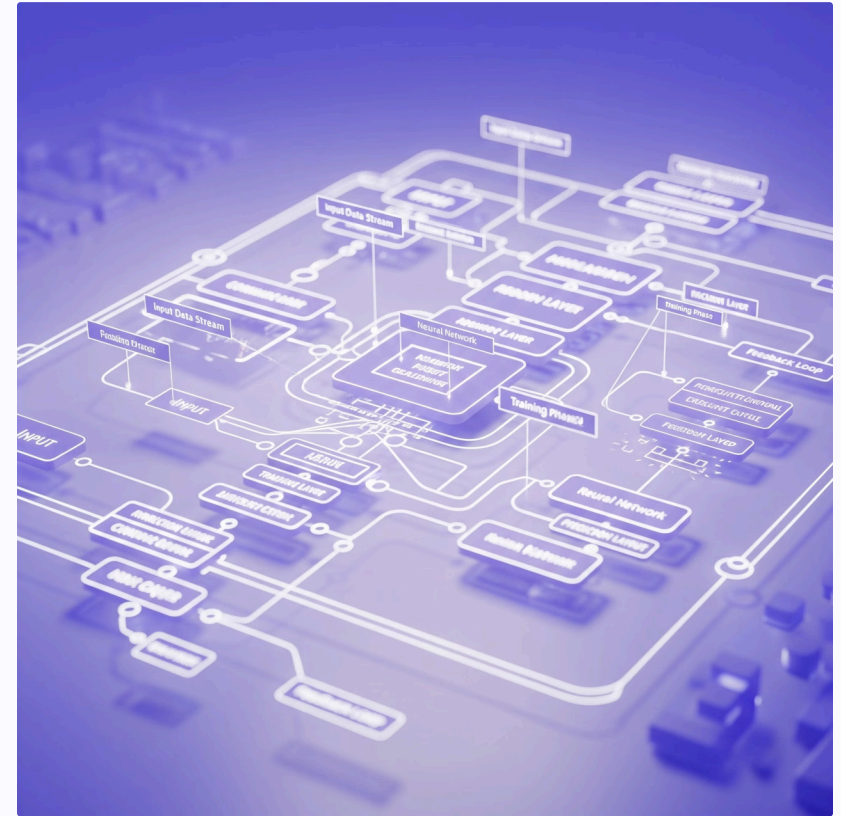
Surfaces critical issues with data availability, compute requirements, and compliance needs before they become blockers

Metric Alignment

Ensures your infrastructure choices directly support your Day 331 metrics and project goals

Stakeholder Communication

Creates a shared visual language to discuss your design with peers, mentors, and evaluators



Core Components of Your Infra Blueprint

1

Data Pipeline

Ingestion mechanisms, preprocessing workflows, feature store implementation, and data versioning strategy

2

Compute Layer

Hardware selection (GPUs/TPUs/NPUs/custom silicon), scaling approach, and resource optimization

3

Training Orchestration

Distributed job management, experiment tracking, hyperparameter tuning, and checkpoint strategy

4

Deployment Layer

Containerization approach, orchestration platform, API design, and serving infrastructure

5

Monitoring + Security

Model drift detection, performance metrics, IAM controls, audit logging, and vulnerability management

Example Blueprint: NLP Service



Data

Text corpus → Tokenization pipeline → Vector DB with version control



Compute

GPU cluster (A100/H100) with auto-scaling for batch training



Training

PyTorch DDP + MLflow tracking with experiment versioning



Deployment

FastAPI container → K8s → Load balancer with replicas



Monitoring

Prometheus (latency) + CodeCarbon (CO₂) + token usage tracking

Example Blueprint: Vision on Edge



Data Pipeline

Camera streams → Real-time preprocessing → Cloud storage with partitioning

Compute Strategy

GPU cluster for training, specialized NPUs for low-power inference at the edge

Training Framework

TensorFlow + Kubeflow pipelines with model compression techniques

Deployment Approach

Quantized models → Edge device containers with failover capability

Monitoring Suite

Grafana dashboards + Device health logs + Bandwidth optimization



Example Blueprint: Generative AI

Data & Compute

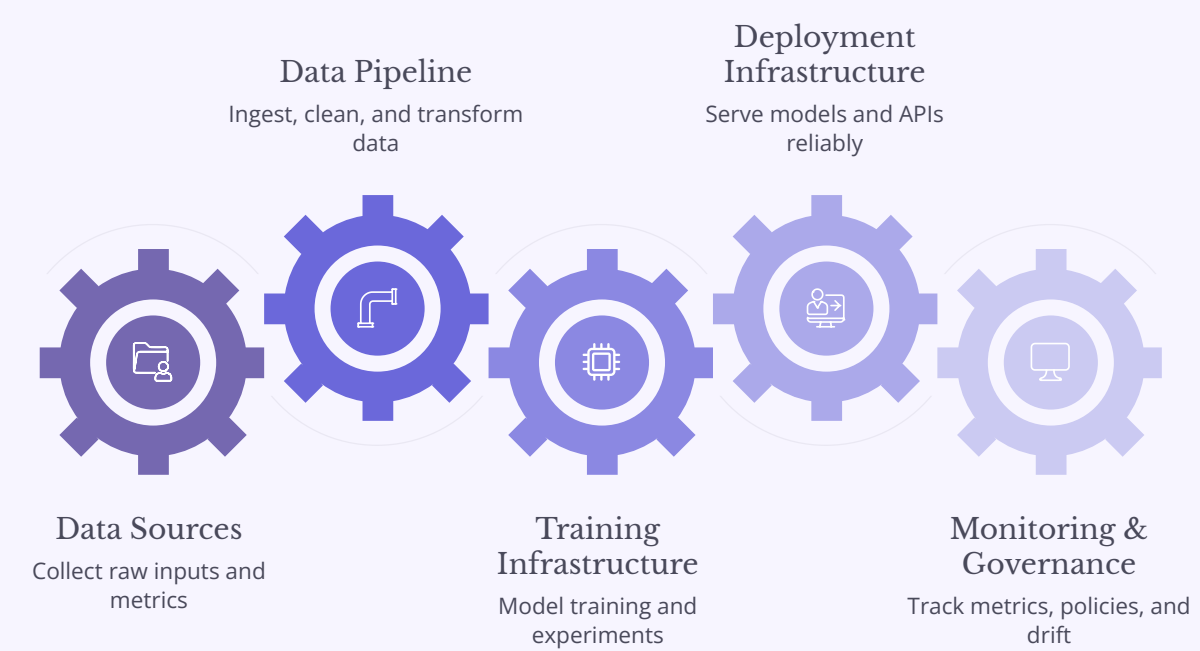
- Large-scale image-text pairs with augmentation pipeline
- Multi-GPU cluster leveraging spot instances for 60% cost reduction
- Gradient checkpointing to manage memory constraints
- DeepSpeed ZeRO sharding for model parallelism

Deployment & Monitoring

- Inference API with request queuing and result caching
- Horizontal pod autoscaling based on request volume
- Comprehensive cost dashboards tracking per-request expenses
- Content safety filters with human-in-the-loop escalation
- Attribution tracking for generated outputs

Blueprint Visual Template

Use this framework to structure your own infrastructure blueprint:



"Blueprint = end-to-end view of AI infrastructure aligned with project metrics"

Adding Compliance & Cost Layers



Your blueprint must go beyond technical components to address:

Compliance Requirements

- Data residency constraints
- Comprehensive audit logging
- Encryption standards (at-rest/in-transit)
- Access control policies

Cost Management

- FinOps dashboards for visibility
- Cost-per-training run tracking
- Resource utilization optimization
- Budget alerts and guardrails

Sustainability Metrics

- CO₂ emissions per inference
- Energy efficiency monitoring
- Hardware lifecycle management

Common Blueprint Mistakes to Avoid



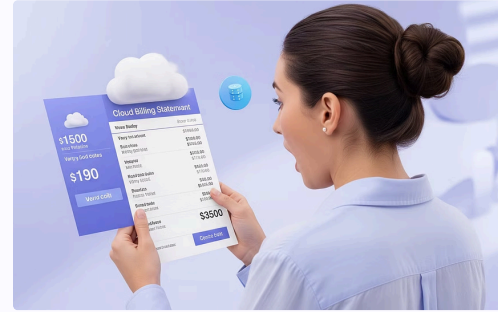
Missing Monitoring Layer

Failing to design robust drift detection and performance tracking from the start, making it impossible to know if your model is performing as expected in production



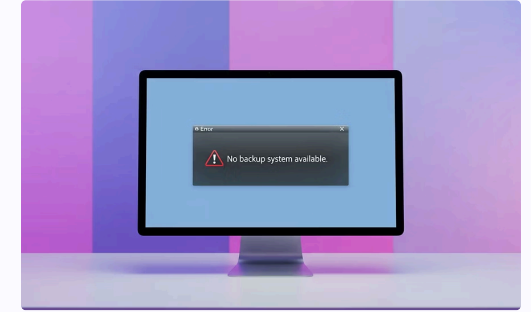
Overdesigning

Creating architecture too complex for your Capstone timeline, leading to implementation delays and incomplete deliverables



Ignoring Cost Trade-offs

Designing without budget constraints, resulting in unsustainable infrastructure that can't be maintained throughout your project timeline



No Fallback Strategy

Lacking contingency plans for hardware/cloud unavailability, creating single points of failure that can derail your entire project

Iterative Design Strategy

Your blueprint will evolve through multiple iterations:



Remember: Iteration > Perfection



Key Lessons for Blueprint Success

Metrics Drive Infrastructure

Your blueprint should clearly show how each infrastructure component supports your specific project metrics and goals

Full Lifecycle Coverage

Design must comprehensively address the entire pipeline: data → compute → deployment → monitoring

Beyond Technical Specs

Incorporate compliance, cost management, and sustainability considerations from the beginning

Embrace Iteration

Prioritize continuous refinement over initial perfection—your blueprint will evolve as your understanding deepens

Discussion: Constraint-Driven Design

"If you had one constraint (budget, latency, or compliance), which would most shape your initial blueprint — and why?"

Budget Constraint

How would limited resources affect your architecture choices? Which components would you prioritize?

Latency Requirement

If real-time performance was critical, how would your infrastructure design change?

Compliance Mandate

What if strict data governance was required? How would this reshape your approach?

Share your constraint-driven blueprint insights with your peers – different constraints often lead to innovative solutions!